

Cybersecurity Incident Report

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log

Incident Summary

After analyzing the tcpdump log, I determined that the incident involved the **Domain Name System (DNS)** service and the **Internet Control Message Protocol (ICMP)**.

The browser attempted to access www.yummyrecipesforme.com by sending a DNS query to the DNS server at IP address **203.0.113.2** using the **User Datagram Protocol (UDP)** on **port 53**, which is the standard port used for DNS services.

Instead of receiving a DNS response containing the website's IP address, the system received an **ICMP error message** stating:

"udp port 53 unreachable"

Protocols Identified

The following protocols were observed in the traffic:

- **DNS** – Used to translate the website domain name into an IP address.
- **UDP** – Used to transport the DNS query.
- **ICMP** – Used to return an error message indicating the DNS request could not be delivered.

Evidence from the Log

- Source IP: **192.51.100.15** (client computer)
- Destination IP: **203.0.113.2** (DNS server)
- DNS query sent for www.yummyrecipesforme.com
- Multiple ICMP responses returned:
 - **"udp port 53 unreachable"**
- The same error occurred repeatedly during multiple attempts.

Interpretation of the Issue

The DNS server could not receive or process requests on UDP port 53. This indicates that:

- The DNS service may be offline.
- Port 53 may be blocked by a firewall.
- The DNS server may not be listening for DNS requests.
- A network configuration issue may be preventing DNS communication.

Because DNS resolution failed, the browser was unable to obtain the website's IP address, preventing access to www.yummyrecipesforme.com.

Part 2: Explain your analysis of the data and provide at least one cause of the incident

When the Problem Was First Reported

The issue was first reported when customers stated they could not access www.yummyrecipesforme.com and received a "**destination port unreachable**" error message.

Scenario, Events, and Symptoms

Several customers reported being unable to load the website.

During troubleshooting:

1. The analyst attempted to access the website.
2. The same error was observed.
3. A packet capture was performed using tcpdump.
4. DNS requests were sent to the DNS server.
5. ICMP error messages were returned indicating:
 - **UDP Port 53 Unreachable**

Current Status

The issue has been identified and escalated to the security engineering team for further investigation and remediation.

The website remains inaccessible because DNS requests are failing before the browser can establish an HTTPS connection to the web server.

Information Discovered During the Investigation

The investigation revealed:

- DNS queries were being transmitted correctly from the client.
- The client was using UDP to contact the DNS server.
- The DNS server responded with ICMP error messages.
- UDP port 53 was unreachable.
- DNS name resolution could not be completed.
- No evidence suggested a problem with HTTPS or the web server itself.

Next Steps

To continue troubleshooting and resolve the issue:

1. Verify that the DNS service is running on the DNS server.
2. Confirm that UDP port 53 is open and listening.
3. Review firewall rules for blocked DNS traffic.
4. Check server logs for DNS service failures.
5. Test DNS functionality from additional devices and networks.
6. Restart the DNS service if necessary.
7. Verify DNS server configuration settings.

Suspected Root Cause

The most likely root cause is that the **DNS service on the DNS server is unavailable or unreachable on UDP port 53.**

Because the DNS server could not process incoming DNS requests, it returned ICMP "port unreachable" messages, preventing domain name resolution and making the website inaccessible to users.

Conclusion

The incident primarily affected the **DNS protocol**, which uses **UDP port 53** for communication. The tcpdump log showed repeated ICMP error messages indicating that UDP port 53 was unreachable. As a result, DNS queries failed, preventing users from resolving the domain name www.yummyrecipesforme.com into an IP address and accessing the website. The most likely cause is a DNS server or port 53 availability issue requiring further investigation by the network and security engineering teams.